

GeoCalib: Supplementary Material

Anonymous submission

Source and Preprocessing Provenance

Open3DSG. The main source-output case study uses the official non-averaged BLIP checkpoint `epoch=13-step=13104.ckpt`. It was selected by train/dev validation loss, not by the held-out GeoCalib result. The documented recovery route covers all 548 evaluation contexts by setting the minimum visible-object count to two and regenerating relaxed views for two scans. The unmodified public-source route covers 533/548 contexts. An older 127-scan workflow covered 377/388 contexts; its corrected comparison covers 388/388 and changes Recall@100 by 0.28 percentage points. These older branches are provenance/sensitivity evidence, not separate main results or candidate selection criteria.

SGFN. We use the official `full_1160` source with its 160-object, 26-relation checkpoint. A pre-execution audit corrected the dataset-list name and an incompatible 120 checkpoint URL. The evaluated scans, denominator, methods, and tensor-shape checks are recorded in the submitted artifacts.

Compatibility Model and Calibration Diagnostics

GeoCalib fits one logistic model per measured relation family. Numeric features are standardized by training-only means and standard deviations; missing numeric values use the corresponding training mean. The input groups are predicate/family indicators T , predicate-independent distances, extents, overlap, contact, and vertical measurements G , and two direction-aligned vertical interactions $T \times G$. Source score and source identity are forbidden. All models use binary cross-entropy, $L_2 = 10^{-4}$, 800 fixed gradient steps, and learning rate 0.2.

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Table 1: **Family calibration on the held-out calibration-dev split.** Labels are the constructed GT-positive/counterfactual target, not human physical-validity judgments.

Family	Train	Dev	Feat.	AUROC	Brier	ECE
Proximity	1,732	382	22	1.0000	.0009	.0120
Vertical order	576	274	23	.9982	.0189	.0437
Support/contact	2,308	537	24	.9831	.0526	.0378

The factor-isolation audit uses the same train/dev rows and estimator class, without inspecting source-result metrics. At dev, T -only, true raw- G -only, additive $T + G$, and interaction $T \times G$ obtain AUROC .6124/.8809/.9193/.9822 and Brier .2370/.1363/.1061/.0495. This establishes that predicate-geometry interaction improves the constructed-target fit; it does not convert the target into independent physical-validity ground truth.

Family Composition at the Global $K = 100$ Operating Point

Table 2 reports each family’s slice inside the actual global top-100 ranking. It therefore exposes both within-family behavior and changes in the family composition of the selected list. Deltas compare the family-calibrated product with the source score, using paired subgraph bootstrap samples.

Table 2: **Global-top-100 family slices.** Positive ΔV is a regression. Values are verifier-derived.

Source	Family	ΔR	ΔV
VL-SAT	Support/contact	+0.0017	+0.0086
VL-SAT	Proximity	+0.0079	−0.0075
VL-SAT	Vertical order	+0.0051	−0.0570
Open3DSG	Support/contact	−0.0523	+0.0059
Open3DSG	Proximity	+0.2135	−0.0462
Open3DSG	Vertical order	+0.1795	−0.2877
SGFN	Support/contact	+0.0011	+0.0086
SGFN	Proximity	+0.0368	−0.0151
SGFN	Vertical order	+0.0128	−0.0602

Support/contact regresses in all three sources, while ver-

tical ordering drives most of the violation reduction. The aggregate result is consequently a scoped operating-point improvement, not a uniform family claim. Full within-family top- K and global-slice tables for $K \in \{5, 10, 20, 50, 100\}$ accompany the submitted artifacts with simultaneous family-wise uncertainty.

Closest-Work Contract Comparison

Table 3 states the novelty boundary directly. GeoCalib does not claim novelty from geometry, calibration, pruning, or rescoring alone.

Table 3: **Closest reliability-adjacent contracts.** “Inside” changes the source model or its training; “post-source” operates on fixed candidate outputs.

Work	Location	Contract and distinction
SCR-SSG (2025)	Inside	Statistical-prior rescoring; no source-excluded same-pair compatibility or verifier-V protocol
RelWitness (2026)	Inside	Witness-guided incomplete supervision, rather than an audit of fixed outputs
SGFormer++ (2026)	Inside	Spatial predicate adaptation; modifies predicate learning
RelGraphOV (2026)	Inside	Geometric pruning for representation learning, not paired recall-V reranking
PUF (2026)	Online	Uncertainty-aware node association, not relation-candidate compatibility
GeoCalib	Post-source	Fixed sources, $Z \notin C$, identity controls, and joint recall/V/uncertainty

Verifier-Uncertainty Sensitivity

The primary $V@K_{all}$ includes uncertain rows in its denominator but does not label them satisfied. We therefore recompute every evaluated ranking with $V@K_{dec}$ over satisfied/violated rows, uncertainty rate $U@K$, and a pessimistic bound $V@K_{pess}$ that counts every uncertain row as a violation. Table 4 gives the primary $K = 100$ comparison.

Table 4: **Uncertainty sensitivity at $K = 100$.** Lower is better for all four reported quantities.

Source	Condition	V_{all}	V_{dec}	U	V_{pess}
VL-SAT	Source score	.0476	.0729	.3464	.3941
VL-SAT	Family product	.0333	.0485	.3127	.3460
Open3DSG	Source score	.1242	.2128	.4164	.5406
Open3DSG	Family product	.0341	.0457	.2539	.2880
SGFN	Source score	.0630	.1005	.3732	.4362
SGFN	Family product	.0381	.0577	.3396	.3777

The paired product-minus-source changes in V_{dec} are $-.0244$ $[-.0270, -.0220]$ for VL-SAT, $-.1671$ $[-.1748, -.1597]$ for Open3DSG, and $-.0428$ $[-.0457, -.0394]$ for SGFN. Uncertainty rate also decreases on all three sources. The corresponding pessimistic-V changes are $-.0480$, $-.2526$, and $-.0586$, with all 95% CIs below zero. Thus the reported improvement does not rely on

promoting uncertain rows. Full K grids and shared-index paired intervals accompany the submitted artifacts.

Cross-Dataset Transfer Development Diagnostic

We evaluate the 11-scene official ReplicaSSG test split with the official VisualGenome-trained FROSS RT-DETR-EGTR release. The initial result is used to diagnose and develop source-scale-robust fusion, so selected configurations are reported as development estimates. The exact mapping is `near`→`close` by, `above`→`higher` than, and `under`→`lower` than; support/contact predicates are excluded because no exact subtype mapping was available. The run contains 4,290 candidates, 172 exact-label ground-truth relations, 11 paired bootstrap units, and passes all 24 source, mapping, and artifact-integrity checks.

Table 5: **Initial ReplicaSSG/FROSS transfer diagnostic at $K = 100$.** Deltas and 95% CIs are paired over the same 1,000 scene-bootstrap samples. The diagnostic criterion requires the recall CI lower bound to exceed -0.01 and the violation CI upper bound to be below zero.

Condition	R	ΔR [95% CI]	V	ΔV [95% CI]
Source score	.3605	—	.1967	—
Family product	.3605	.0000 [.0000,.0000]	.1967	.0000 [.0000,.0000]
Rank-average	.3314	-.0291 [-.0741,.0133]	.0384	-.1583 [-.1929, -.1219]
RRF ($c = 60$)	.3314	-.0291 [-.0579, -.0051]	.0595	-.1372 [-.1682, -.1045]

Neither initial GeoCalib instantiation passes. Product is identical to semantic ranking at the primary endpoint, so it fails the strict violation-improvement criterion. Rank-average sharply lowers verifier-derived V but fails the recall guardrail. Product improves secondary $K = 20$ and $K = 50$ tradeoffs, but these do not replace the primary $K = 100$ result. This failure exposes raw-product score-scale sensitivity, context saturation at $K = 100$, excessive displacement under rank fusion, geometry-representation shift, and a highly imbalanced near/above/under ontology. We use these observations for method development and do not interpret this table as an unbiased dataset-generalization estimate. V remains verifier-derived rather than independent human physical validity.

Bounded-fusion follow-up. The regenerated FROSS execution contains 4,293 candidates, three more than the initial execution; every condition below uses this same regenerated candidate set. We evaluate 355 context-quantile and bounded-penalty configurations, including decoders constrained by $|\text{rank}_{new} - \text{rank}_{source}| \leq B$ for $B \in \{10, 20, 50, 100\}$. The all-scene choice $q^Z - [.6 - C]_+$ preserves Recall@100 and lowers V, but it is an optimistic development fit. Selecting on ten scenes and evaluating the held-out scene loses recall and fails the diagnostic criterion.

Table 6: **ReplicaSSG bounded-fusion development estimates at $K = 100$.** The LOSO row is the relevant internal estimate; the all-scene row is optimistic.

Condition	R	ΔR	V	ΔV
Source score	.3547	–	.1967	–
Family product	.3547	.0000	.1967	.0000
Rank-average	.3256	–.0291	.0422	–.1545
Bounded, all-scene	.3547	.0000	.0393	–.1574
Bounded, LOSO	.3198	–.0349	.0384	–.1583

For LOSO, the paired 95% intervals are $[-.0755, .0000]$ for ΔRecall and $[-.1955, -.1161]$ for $\Delta\text{Violation}$. Thus bounded demotion is a useful development direction but does not repair the cross-dataset generalization limitation, and it is not part of the main claim.